Quiz I

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

A cache is primarily used to improve:

Select Multiple Options

Correct
☑ Latency

False
☐ Concurrency

False
☐ Fault tolerance

False
☐ Cost

Check Answer

Correct Answer ✓

Quiz ?

Statement

In a distributed environment, a network partition can cause which of the following problems? Select all that apply.

Select Multiple Options

Correct
☑ Split-brain

Correct
☑ Each of the two servers might think that the other one is dead.

False
☐ Messages might be sent to the wrong server.

Correct
☑ Message or data loss.

Explanation
E.g., in Dynamo, when a node is down, its data is written to an intermediate node, and if that intermediate node dies before transferring the data to the correct node, the data will be lost.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the benefit of using Piggybacked acknowledgments?

Select Multiple Options

False
☐ It delays the acknowledgment response.

Correct
☑ It efficiently uses the available channel bandwidth, thus improving the efficiency of the system.

Explanation
Piggybacking saves bandwidth as it removes the need of sending a separate acknowledgment message.

False
☐ It decreases the system complexity.

Explanation
Actually, Piggybacking increases the system's complexity.

False
☐ It is an efficient way for the sender to acknowledge the receipt of an acknowledgment.

Check Answer

Correct Answer ✓

Quiz ?

Statement

The CAP theorem tells us that:

Select Multiple Options

False
☐ A distributed system can only have two out of three. Consistency, atomicity, and partition tolerance.

Explanation
CAP theorem does not talk about 'atomicity'.

False
☐ A distributed transaction must have to choose two out the three. Consistency, isolation, and durability.

Correct
☑ Any distributed system which is highly-available will have problems with consistency.

Explanation
CAP theorem can be summarized as, "In the presence of a network partition, a distributed system must choose either Consistency or Availability."

False
☐ In distributed systems, durability and lower latency can only come at the cost of lower consistency.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Following is the property of the Write-ahead Log (WAL). Select all that apply.

Select Multiple Options

Correct
☑ Data is always sequentially appended to the WAL.

Correct
☑ Each log entry in the WAL has a unique identifier.

Correct
☑ WAL does not allow updating the data once written.

False
☐ WAL reduces the read latency by providing a cache containing recent transactions.

Check Answer

Correct Answer ✓

Quiz ?

Statement

As opposed to locks, leases:

Select Multiple Options

Correct
☑ Have expiration times.

Explanation
Leases are just locks that expire.

False
☐ Are long-term while locks are short-term

False
☐ Can be shared between multiple clients

False
☐ Are coarse-grained locks

Check Answer

Correct Answer ✓

Quiz ?

Statement

What advantage does it give to a system to have separate read and write locks?

Select Multiple Options

Correct
☑ Enables concurrent reading of a resource by multiple entities

Explanation
Since read locks keep out writers, multiple readers can access a resource as no one will be modifying it.

False
☐ Enables concurrent writing to a resource by multiple entities

False
☐ Enables entities to get 'execute' access on a resource if they acquire both read and write locks at the same time

False
☐ None of the above

Check Answer

Correct Answer ✓

Quiz ?

Statement

For GFS, what are the benefits of a large chunk size? Select all that apply.

Select Multiple Options

False
☐ A small file can all be stored in just one chunk.

Correct
☑ Large chunk size provides highly efficient sequential reads and appends of large amounts of data.

Correct
☑ A large chunk size simplifies ChunkServer management.

Explanation
I.e., to check which ChunkServers are near capacity or which are overloaded.

Correct
☑ A large chunk size reduces the size of the metadata stored on the master.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Consistent hashing ensures that:

Select Multiple Options

False
☐ Data will be stored on the servers reliably.

Correct
☑ If a server is removed, its data will be distributed to all other nodes in the cluster.

Correct
☑ Only a small set of keys move when servers are added or removed.

Correct
☑ The data will be distributed equally among the servers.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Dynamo uses virtual nodes for which of the following reasons? Select all that apply.

Select Multiple Options

Correct
☑ To avoid hotspots

Correct
☑ To spread the load more evenly across the physical nodes on the cluster by dividing the hash ranges into smaller subranges

False
☐ To enable the use of virtual machines and virtual memory

Correct
☑ To better maintain a cluster containing heterogeneous machines

Check Answer

Correct Answer ✓

Quiz ?

Statement

Of the following options, which is not a characteristic of Amazon's Dynamo?

Select Multiple Options

False
☐ Partition-tolerant

False
☐ Highly available

Correct
☑ Strongly Consistent

False
☐ Highly reliable

Check Answer

Correct Answer ✓

Quiz ?

Statement

Unlike BigTable, with Amazon Dynamo:

Select Multiple Options

False
☐ Keys are sorted alphabetically to support iteration.

False
☐ An object is identified by exactly one key.

Correct
☑ Two processes may write conflicting updates.

False
☐ All requests pass through a coordinator.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Virtual nodes in Amazon Dynamo are designed to:

Select Multiple Options

False
☐ Improve fault tolerance due to the replication of nodes

False
☐ Increase the requests the system can handle since many virtual nodes can be managed by one physical node

Correct
☑ Improve load distribution when adding or removing nodes

False
☐ Create an overlay network that arranges nodes into a logical ring

Check Answer

Correct Answer ✓

Quiz ?

Statement

Which of the following are in-memory structures where Cassandra buffers write?

Select Multiple Options

False
☐ SSTables

False
☐ NetTables

False
☐ LinkTables

Correct
☑ Memtables

Check Answer

Correct Answer ✓

Quiz ?

Statement

What are the disadvantages of using Tombstones in Cassandra? Select all that apply.

Select Multiple Options

False
☐ Slower reads

False
☐ Deleted data takes more storage until Tombstones are completely removed during compaction.

Correct
☑ Slower writes

False
☐ Faster writes

Check Answer

Correct Answer ✓

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Quiz II

☐ Mark as Completed